

A Privacy-Respecting User Activity Classification Method Based on O-RAN E2 Metrics

Ioannis Ktenidis*, Athanasios Liatifis*, Dimitrios Pliatsios*, Evangelos Markakis[‡],
Panagiotis Fouliras[§], Vasileios Vitsas[¶], Thomas Lagkas^{†*}, Panagiotis Sarigiannidis*

*Department of Electrical and Computer Engineering,
University of Western Macedonia, Kozani, Greece

e-mails: {aff02156, aliatifis, dpliatsios, psarigiannidis}@uowm.gr

[‡]Department of Electrical & Computer Engineering,
Hellenic Mediterranean University, Heraklion, Greece
e-mail: emarkakis@hmu.gr

[§]Department of Applied Informatics, University of Macedonia, Thessaloniki, Greece
e-mail: pfoul@uom.edu.gr

[¶]Department of Information and Electronic Systems Engineering,
International Hellenic University, Thessaloniki, Greece
email: vitsas@it.teithe.gr

[†]Department of Informatics, Democritus University of Thrace,
Kavala Campus, Kavala, Greece
e-mail: tlagkas@cs.duth.gr

Abstract—The emergence of fifth-generation (5G) networks and the Open Radio Access Network (O-RAN) paradigm aspire to offer more flexible, programmable, and intelligent mobile infrastructures. A key enabler for autonomous resource orchestration in such networks is the accurate and privacy-respecting classification of user activity from RAN telemetry. In this paper, we present the design, deployment, and evaluation of an O-RAN-compliant 5G testbed used to collect near-real-time metrics via the E2 interface. We apply machine learning (ML) techniques on these measurements to classify user activity into three service categories, namely gaming, video streaming, and browsing. Specifically, a deep neural network (DNN) and an extreme gradient boosting (XGBoost) model were trained to classify the user activity into the aforementioned categories. The evaluation results show a high overall performance for both DNN and XGBoost. Specifically, the precision, recall, F1-score, and accuracy values of the DNN are 0.9208, 0.9203, 0.9204, and 0.9204, respectively. On the other hand, the corresponding values for the XGBoost are 0.9801, 0.9794, 0.9791, and 0.9794. The results show that the combination of metrics from the E2 interface and ML can offer insights into the user activity, which are valuable for optimal resource allocation, without having to directly inspect the network packets.

Index Terms—B5G, Classification, Deep Neural Network, E2 Metrics, eXtreme Gradient Boosting (XGB), Open-Radio Access Network, Software Defined Radio, xApp

I. INTRODUCTION

The emergence of next-generation mobile networks aims to offer increased data rates, ultra-low latency, and massive connectivity [1]. One significant challenge in realizing the full potential of next-generation mobile networks is the efficient orchestration of network resources in dynamic environments.

To this end, the classification of user activity aims to identify the type of applications or services a user is engaged in. The insights into the user's activity can be leveraged for the efficient orchestration of wireless resources.

Traditional methods for identifying user activity often rely on deep packet inspection (DPI) or statistical analysis of network traffic flows [2]. The authors of [3] developed a DPI method based on ensemble learning to investigate user activities in the network. Moreover, the authors of [4] analyzed voice traffic in 5G using long-short-term memory and convolutional neural network models. The results showed that the long-short-term memory model had better performance for Over-The-Top voice traffic. In [5], the authors used a deep neural network to identify user activities in mobile applications from encrypted network traffic. A similar machine learning-based approach was presented by Shahraki *et al.* in [6], which also highlighted the associated challenges of applying machine learning in identifying the user activity. In [7], the authors designed a deep learning model to classify the user activity from encrypted network traffic. The methods utilized in the aforementioned works can raise privacy concerns, while they may not be flexible enough to adapt to the requirements of next-generation networks. Moreover, the work in [8] investigated the use ML-based classification to predict the load in a 5G network and allocate the respective resources, while the work in [9] used ML for packet inspection in order to detect and identify enhanced mobile broadband (eMBB) activities. Finally, the authors of [10] developed a convolutional neural network

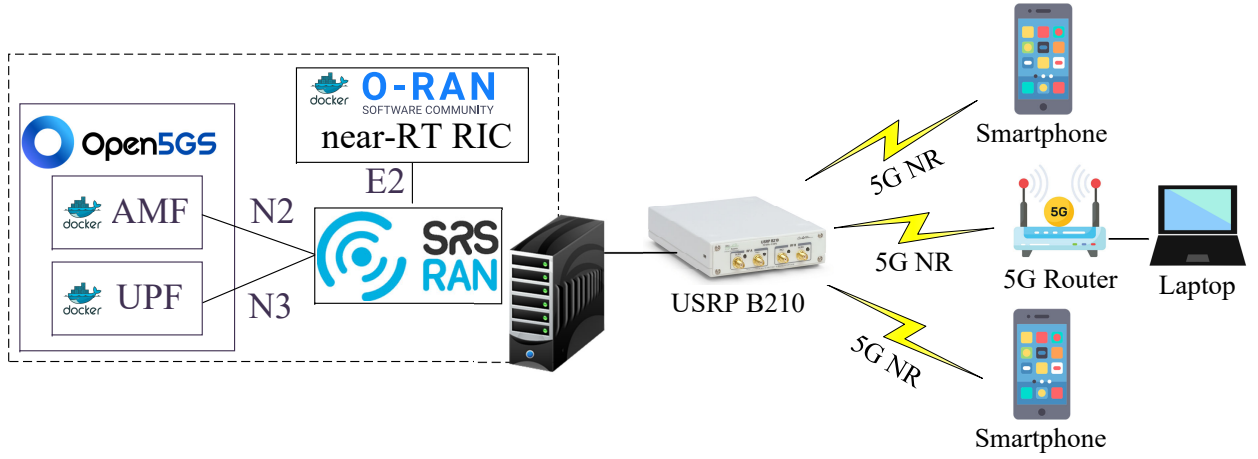


Fig. 1. Testbed Architecture & Topology

(CNN) to classify between various user activities using data from the Open Radio Access Network (O-RAN) E2 interface.

Unlike most works, we present an activity classification approach that takes advantage of the network metrics available from the E2 interface. Compared to the work presented in [10], we deploy a local O-RAN 5G testbed and generate the dataset using real end-user devices. Also, we implement and compare two different ML-based methods, namely Deep Neural Network and XGBoost.

O-RAN is a new paradigm that aims to foster innovation, reduce manufacturer lock-in, increase interoperability, and promote a flexible and programmable network infrastructure [11]. It enables the integration of intelligent methods for the orchestration of network resources through the deployment of xApps. Using artificial intelligence (AI)/machine learning (ML), these applications can improve network performance, resource allocation, and quality of service (QoS) [12]. In more detail, the E2 interface enables the exchange of near-real-time control and management information between the O-RAN radio unit (O-RU), distributed unit (O-DU), centralized unit (O-CU), and the RAN intelligent controller (RIC) [13]. The information obtained through the E2 interface can be analyzed to gain insights into network's state and users' behavior without resorting to DPI-based methods.

In this work, we present the design and deployment of an O-RAN-compliant 5G testbed, which is used to collect data from the E2 interface. The data were used to train two classifiers, namely a deep neural network (DNN) and an extreme gradient boost (XGBoost) model. This work's contributions are highlighted as follows:

- We present the deployment of the O-RAN-compliant 5G testbed, the main interfaces, and the utilized hardware and open source software components.
- We describe the process for collecting the data from the E2 interface and structuring the respective dataset. For the data generation, three user activities were carried out, namely gaming, video streaming, and browsing.

- We present the design and main principles of the DNN and the XGBoost model. These models were trained through the generated dataset using a 70/30 split.
- We evaluate the performance of two classification methods, namely the DNN and the XGBoost, in terms of precision, recall, F1-score, and accuracy.

The outcomes of this work aim to demonstrate the feasibility of utilizing data from O-RAN E2 interface in classifying the user's activity in a privacy-respecting manner. The user activity classification has the potential to enhance the orchestration of resources toward improving the network performance. Contrary to DPI-based methods, the proposed approach has the potential to foster the consumers' trust to the network operator, as the analysis of the consumer traffic is not required.

The rest of the paper is organized as follows: Section II describes the testbed and dataset generation process, while Section III analyzes the two ML models. Furthermore, Section IV presents the performance evaluation results. Finally, Section V summarizes and concludes this paper.

II. TESTBED DEPLOYMENT & DATASET GENERATION

A. Testbed Architecture, Hardware, and Software

The architecture of the testbed is illustrated in Fig. 1. The testbed consists of three hardware devices, namely a general-purpose computing server, an Ettus Research USRP B210 device, two smartphones, and a 5G router, which equips a laptop with 5G capabilities.

A general-purpose computing server is used to host the necessary software for deploying the 5G new radio (NR) base station (BS). In more detail, the Open5GS software is used to provide core network (CN) functionalities. Specifically, the access and mobility management function (AMF) is responsible for handling the registration, connection, and mobility management of the devices, while the user plane function (UPF) is responsible for carrying the user data packets between the gNB and the external network [14]. Additionally, the srsRAN software is responsible for managing the USRP B210 device and

TABLE I
LIST OF E2 METRICS INCLUDED IN THE DATASET

ID	Metric	Description
F1	DRB.RlcSduTransmittedVolumeDL	Volume of data that are transmitted in the downlink
F2	DRB.UETHPDL	Average throughput of the UE in the downlink
F3	DRB.RlcSduTransmittedVolumeUL	Volume of data that are transmitted in the uplink
F4	RRU.PrbAvailUL	Average number of PRBs that are available in the uplink
F5	RRU.PrbTotUL	Percentage of utilized PRBs in the uplink
F6	DRB.PacketSuccessRateUlgNBUu	Percentage of packet data convergence protocol (PDCP) packets that were successfully received
F7	DRB.RlcSduDelayDL	Average delay experienced by RLC Service Data Units (SDUs) in the downlink.
F8	DRB.RlcPacketDropRateDL	Percentage of the radio link control packets that are dropped in the downlink channel due to high traffic load
F9	RRU.PrbAvailDL	Average number of physical resource blocks (PRBs) that are available in the downlink
F10	RRU.PrbTotDL	Percentage of utilized PRBs in the downlink
F11	DRB.UETHPUL	Average throughput of the UE in the uplink
F12	DRB.AirIfDelayUL	Average over-the-air packet delay in the uplink

deploying the 5G network [15]. srsRAN uses the N2 and N3 interfaces to communicate with the AMF and UPF, respectively. Finally, the near-real-time RIC (near-RT RIC) from O-RAN Software Community [16] is connected with srsRAN through the E2 interface and is responsible for collecting network metrics and sending control messages in a near-RT timescale. The analysis of the metrics and the generation of the appropriate control messages are implemented as xApps, which run in the near-RT RIC. It should be noted that the Open5GS functions and the near-RT RIC are deployed as Docker containers, further enhancing the deployment flexibility.

The Ettus Research USRP B210 undertakes the role of the radio unit. It is a highly versatile single-board platform designed for software defined radio (SDR) experimentation [17]. It offers continuous frequency coverage from 70 MHz to 6 GHz, making it suitable for a wide range of applications, including FM radio, cellular, and WiFi communications. Finally, for the user equipment (UE), a commercial-off-the-shelf (COTS) smartphone running the Android operating system is utilized. The wireless configuration parameters are summarized in Table II. The n77 5G NR band, with a frequency of 4050 MHz, is used for both the downlink (DL) and uplink (UL), combined with time division duplexing (TDD). Moreover, the total channel bandwidth and subcarrier spacing are 40 MHz and 30 kHz, respectively. Finally, the 256-quadrature amplitude modulation (QAM) is used for symbol modulation, while a single-input single-output (SISO) antenna configuration is adopted.

TABLE II
WIRELESS CONFIGURATION PARAMETERS

Parameter	Value
5G NR Band	n77
Frequency	4050 MHz
Duplexing	TDD
Bandwidth	40 MHz
Subcarrier Spacing	30 kHz
Modulation	256-QAM
Antenna Configuration	SISO

B. Dataset Generation

To generate the dataset, each user was engaged in several activities, namely web browsing, gaming, and video streaming. Moreover, an xApp was developed that collects the metrics provided by the srsRAN to the E2 interface and structures them in a comma-separated values (CSV) file. E2 is core interface in the O-RAN architecture that connects the Near-RT RIC with the radio units. The E2 interface enables the collection of telemetry and performance measurements from the radio units and the control of various parameters for efficient resource management. This approach enables third-parties to develop xApps, without the need to modify the underlying protocol stack.

The recorded metrics and their respective descriptions are listed in Table I. As the users engaged in the activities, the aforementioned metrics were recorded every 1 second, along

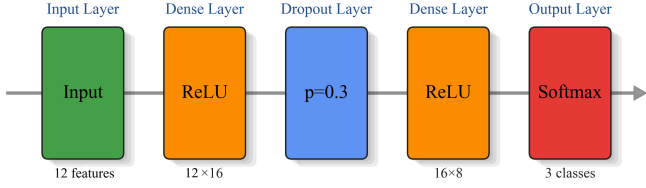


Fig. 2. Deep Neural Network Architecture

with the respective timestamp. Afterwards, the records were manually annotated by adding a label class for each type of activity. Additionally, a data cleaning process, focusing on removing outliers and/or missing values, was applied. The resulting dataset had 1692 entries (564 entries for each class), with each entry having 12 features and one label field.

III. MACHINE LEARNING MODEL

We use time-series E2 interface metrics to train and evaluate two classifiers, namely a DNN and an XGBoost model. The original dataset is preprocessed to address any missing values or extreme outliers using Robust scaling. Next, the dataset is split with a stratified 70/30 train/test split. From the training split, 12.5% is held out as a validation set for early stopping and model selection.

A. Deep Neural Network Classifier

The architecture of the DNN is shown in Fig. 2. In more detail, it consists of five layers, namely the Input layer, two Dense layers, the Dropout layer, and the Output layer. The input layer has $M = 12$ features, which correspond to the number of E2 metrics (Table I). Next, the two dense layers have 16 and 8 neurons, respectively, while each dense layer uses the ReLU activation. To mitigate overfitting, a dropout layer with a rate of $p = 0.3$ follows the first dense layer. Finally, the output layer contains $C = 3$ neurons corresponding to the number of classes (i.e., gaming, streaming, browsing). The final prediction is generated through the Softmax activation function:

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}, \quad i = 1, \dots, C \quad (1)$$

where z_j stands for the unnormalized score the model produces for class j before the softmax function.

The Adam optimizer with an initial learning rate of 10^{-3} is utilized. Furthermore, the StepLR scheduler reduces the learning rate by a factor of 0.5 every 100 epochs to ensure stable convergence during training. We use the cross-entropy loss function as the training objective. The training phase runs for up to 1400 epochs, but training can terminate earlier using an early stopping criterion, especially in scenarios with stringent latency constraints. Specifically, after each epoch, the validation cost is calculated, and if the validation loss does not improve by more than 10^{-4} , the training is stopped to prevent further overfitting.

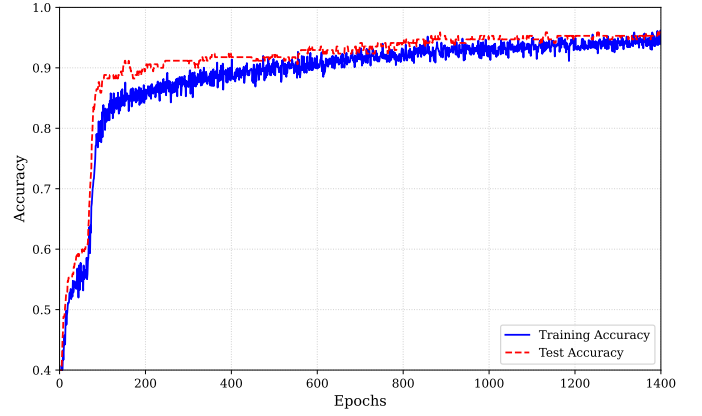


Fig. 3. Accuracy of the Model over the Training Epochs

B. XGBoost Classifier

The XGBoost [18] is configured for multi-class classification using the `multi:softprob` objective with three output classes respectively for gaming, streaming, browsing. Specifically, the XGBoost Tree Booster is selected, while the tree's max depth is 6. Moreover, the number of estimators is set to 100, and the subsample ratio is set to 0.8 to prevent overfitting. Hyperparameter tuning is performed using a grid search with 5-fold cross-validation over the number of estimators, tree depth, learning rate, and subsampling parameters. The final model uses the best-performing configuration identified during this search. The multi-class log loss (`mlogloss`) function used, while feature importance is later computed using the tree split method. Finally, the optimiser is based on gradient boosting, while the learning rate is 0.01.

IV. EVALUATION

In this section, we present the evaluation of the DNN and the XGBoost model. Specifically, four metrics are used to assess the performance of both models, namely precision, recall, F1-score, and accuracy.

A. Deep Neural Network Classifier

Fig. 3 illustrates the training and validation accuracy of the DNN model over the training process. The curves initiate from an accuracy of approximately 0.42 and exhibit a consistent upward trend, converging at approximately 0.92 after 900 epochs. A key observation is the minimal generalization gap between the training and validation curves, which indicates that the model generalizes effectively to unseen data without exhibiting signs of overfitting.

Furthermore, Fig. 4 shows the train and test loss of the DNN over 1400 epochs. A significant and sharp decline is observed during the first 200 epochs, where the loss drops toward the 10^{-1} scale, indicating that the Adam optimizer efficiently tunes the model parameters in the early stages. Notably, the test loss closely tracks the training loss without significant fluctuations, confirming a stable generalization and the absence of overfitting. The convergence of both curves

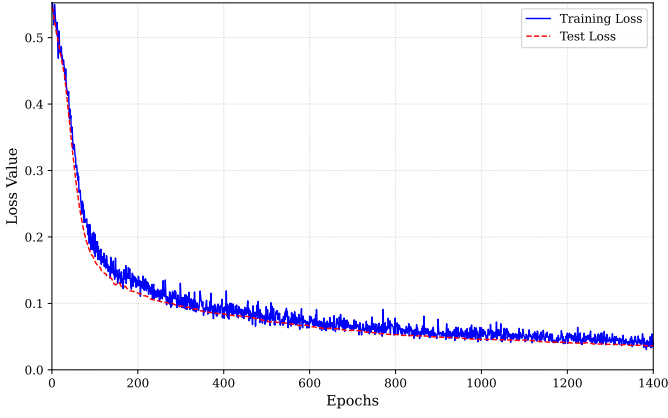


Fig. 4. Loss of the Model over the Training Epochs

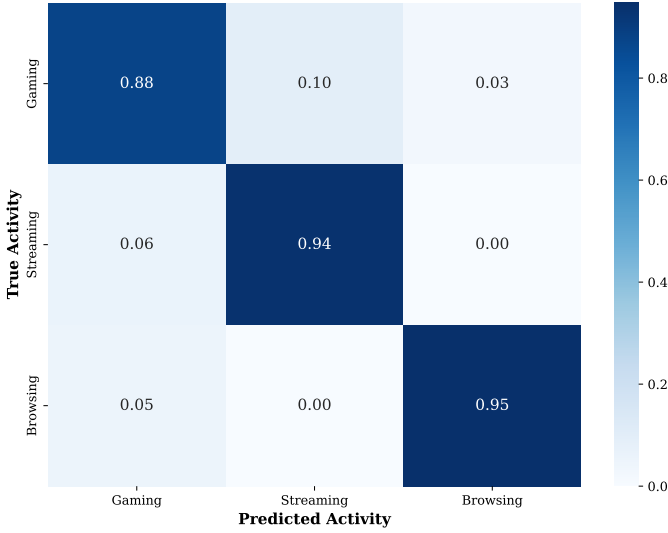


Fig. 5. Normalized Confusion Matrix

at very low values (approximately 0.035) demonstrates the robustness of the DNN in distinguishing between Gaming, Streaming, and Browsing activities. The normalized confusion matrix of the DNN-based classifier is illustrated in Fig. 5. The model achieves high classification performance, with diagonal values of 0.88, 0.94, and 0.95, for gaming, streaming, and browsing, respectively. A small number of misclassifications is noticed between gaming and browsing. This suggests that there are some overlaps in their respective E2 metric patterns.

Fig. 6 shows the precision-recall results for the three activity classes. The method achieves average precision scores of 0.9269 for gaming, 0.9758 for streaming, and 0.9885 for browsing. The results indicate that the model maintains high precision across a wide range of recall values. This means that the DNN classifier can accurately distinguish the user activities, even in challenging scenarios.

Finally, Table III lists the performance values of the DNN classifier, in terms of precision, recall, F1-score and accuracy. Also, the table lists the precision, recall, and F1-score per-

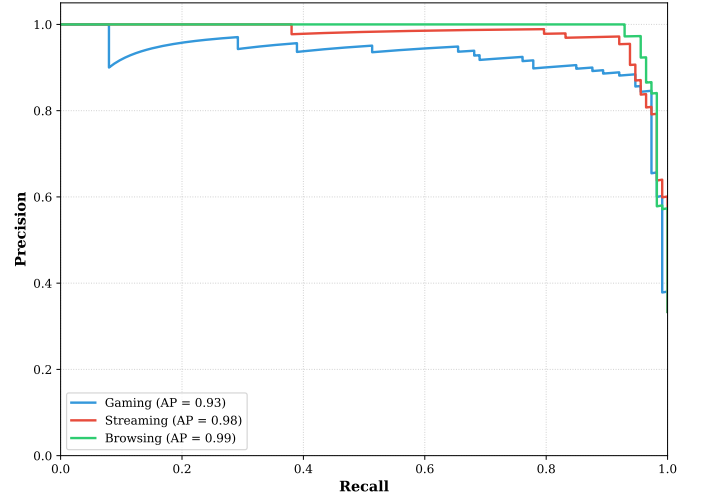


Fig. 6. DNN Precision compared to Recall

TABLE III
DNN CLASSIFICATION PERFORMANCE

Class	Precision	Recall	F1-Score	Accuracy
Overall	0.9208	0.9203	0.9204	
Gaming	0.8839	0.8761	0.8800	0.9204
Streaming	0.9060	0.9381	0.9217	
Browsing	0.9727	0.9469	0.9596	

mance for each particular user activity class. The aforementioned metrics feature high values for all classes, indicating that the DNN model generalizes well across all classes.

B. XGBoost Classifier

Fig. 7 presents the importance of the features in the XGBoost-based classification process. The twelve features included in the dataset are ranked, with scores representing the cumulative contribution of each metric across all decision trees. The scores are calculated based on the tree split method of

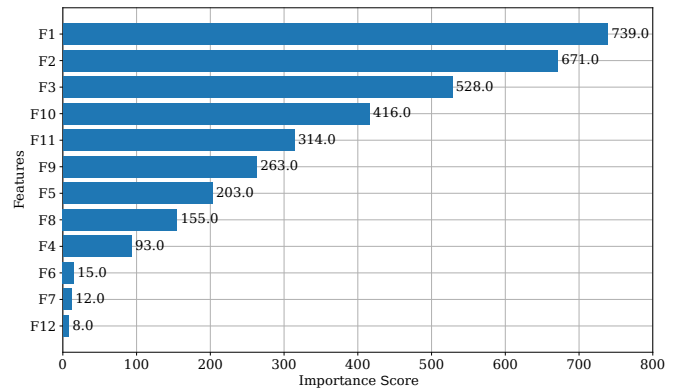


Fig. 7. XGBoost Feature Importance

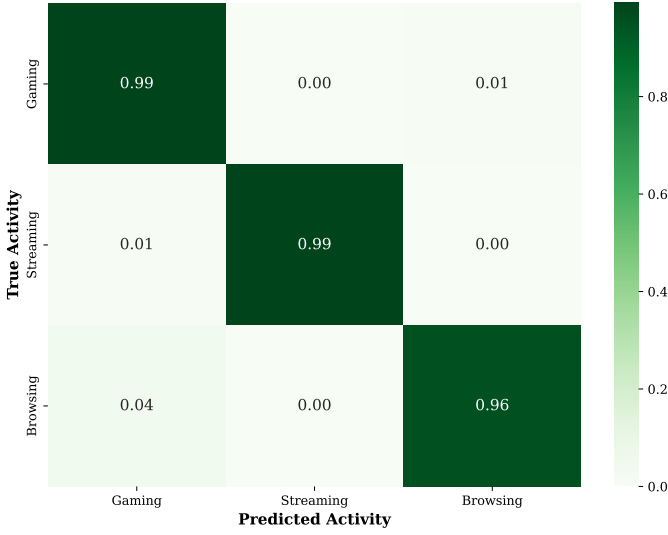


Fig. 8. XGBoost Confusion Matrix

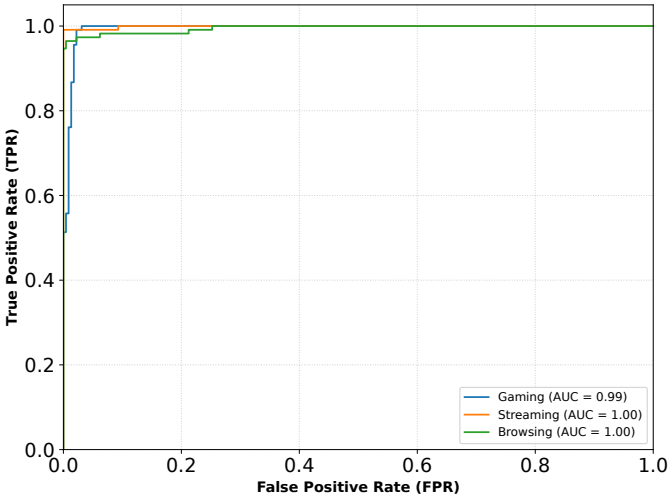


Fig. 9. XGBoost Receiver Operating Characteristic

XGBoost, while the mapping of the feature numbers to E2 metrics is presented in Table I.

The normalized confusion matrix of the XGBoost classifier is illustrated in Fig. 8. As in the DNN case, XGBoost achieves high classification performance, with diagonal values of 0.99, 0.99, and 0.96, for gaming, streaming, and browsing, respectively. Moreover, the 0.04 misclassification error between browsing and gaming indicates a partial overlap in the respective E2 metrics patterns.

Fig. 9 presents the receiver operating characteristic (ROC) results for the XGBoost model. All three classes achieve high separability, with area under curve (AUC) scores above 0.99. Notably, the streaming and browsing class achieves an AUC of 1.00, indicating near-perfect classification.

Moreover, Fig. 10 illustrates the precision–recall results of the XGBoost classifier for all activity classes. The streaming

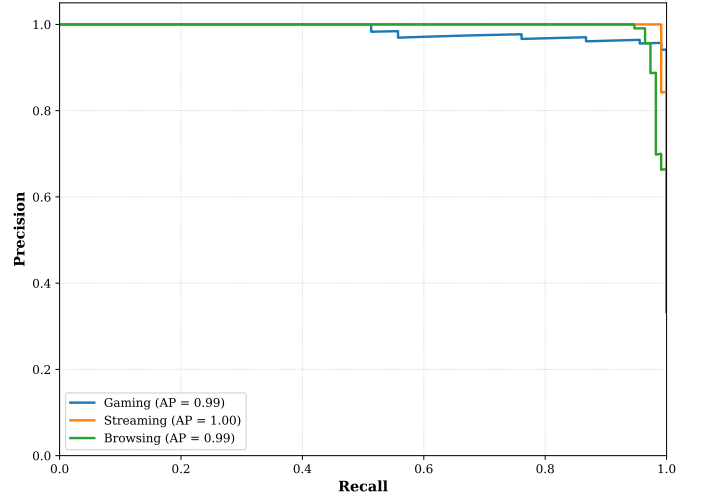


Fig. 10. XGBoost Precision Compared to Recall

TABLE IV
XGBOOST CLASSIFICATION PERFORMANCE

Class	Precision	Recall	F1-Score	Accuracy
Overall	0.9801	0.9794	0.9791	
Gaming	0.9492	0.9912	0.9697	0.9794
Streaming	1.0000	0.9912	0.9956	
Browsing	0.9908	0.9558	0.9730	

class achieves a perfect precision-recall tradeoff (1.00), while the gaming and browsing classes maintained high precision values of 0.99, respectively. The results confirm that the XGBoost model is able to maintain high precision performance across a various recall thresholds, ensuring accurate classification.

Finally, Table IV lists the performance values of the XGBoost classifier, in terms of precision, recall, F1-score and accuracy. Also, the table lists the precision, recall, and F1-score performance for each particular user activity class. The aforementioned metrics feature high values for all classes, indicating that the DNN model generalizes well across all classes. Concerning the prediction time-frame, the tests showed that after 10 seconds, the performance improvement was marginal. Finally, the prediction performance can be further improved by using a more granular interval for recording the E2 metrics.

V. CONCLUSION

In this work, we presented a novel privacy-respecting methodology for classifying user activity by leveraging data from the O-RAN E2 interface, without relying on DPI-based methods. Specifically, we presented the design and deployment of an O-RAN-compliant 5G testbed, which was used to generate and collect the E2 metrics. Using the collected data, we trained and evaluated a DNN and an XGBoost model for classifying user activities into three distinct categories, namely web browsing, gaming, and video streaming.

The results show a high overall performance for both DNN and XGBoost. Specifically, the precision, recall, F1-score and accuracy values of the DNN are 0.9208, 0.9203, 0.9204, and 0.9204, respectively. On the other hand, the corresponding values for the XGBoost are 0.9801, 0.9794, 0.9791, and 0.9794. The results demonstrate the feasibility and effectiveness of utilizing O-RAN interface data to gain deep insights into user behavior, thereby enabling privacy-respecting resource management in mobile networks.

As future work, we aim to develop an xApp that will allocate the radio resources based on the user activities, prioritizing the users that have larger demands in terms of network capacity. In this direction, the near-RT-RIC will be used for retrieving the metrics and enforcing the corresponding allocation decisions. Furthermore, the model will be extended to distinguish between additional user activity classes and applications.

ACKNOWLEDGMENT

This project has received funding from the National Recovery and Resilience Plan ‘Greece 2.0’ under the European’s Union’s NextGenerationEU framework (ASKOS - grant agreement no. YP3TA-0560864).

REFERENCES

- [1] D. Pliatsios, P. Sarigiannidis, S. Goudos, and G. K. Karagiannidis, “Realizing 5G vision through cloud RAN: technologies, challenges, and trends,” *EURASIP Journal on Wireless Communications and Networking*, vol. 2018, no. 1, p. 136, May 2018.
- [2] V. Labayen, E. Magaña, D. Morató, and M. Izal, “Online classification of user activities using machine learning on network traffic,” *Computer Networks*, vol. 181, p. 107557, 2020.
- [3] M. Nkongolo, J. P. van Deventer, and S. M. Kasongo, “Using deep packet inspection data to examine subscribers on the network,” *Procedia Computer Science*, vol. 215, pp. 182–191, 2022.
- [4] Z. Qiao, L. Zhai, S. Zhang, and X. Zhang, “Encrypted 5G over-the-top voice traffic identification based on deep learning,” in *2021 IEEE Symposium on Computers and Communications (ISCC)*, 2021, pp. 1–7.
- [5] M. H. Pathmaperuma, Y. Rahulamathavan, S. Dogan, and A. M. Kondo, “Deep learning for encrypted traffic classification and unknown data detection,” *Sensors*, vol. 22, no. 19, 2022.
- [6] A. Shahraki, M. Abbasi, A. Taherkordi, and A. D. Jurcut, “Active learning for network traffic classification: A technical study,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 8, no. 1, pp. 422–439, 2022.
- [7] X. Han, G. Xu, M. Zhang, Z. Yang, Z. Yu, W. Huang, and C. Meng, “DE-GNN: dual embedding with graph neural network for fine-grained encrypted traffic classification,” *Computer Networks*, vol. 245, p. 110372, 2024.
- [8] T. Tsourdinis, I. Chatzistefanidis, N. Makris, T. Korakis, N. Nikaein, and S. Fdida, “Service-aware real-time slicing for virtualized beyond 5g networks,” *Computer Networks*, vol. 247, p. 110445, 2024.
- [9] J. N. M. Dahj, K. A. Ogudo, and L. Boonzaaier, “A hybrid analytical concept to qoe index evaluation: Enhancing embb service detection in 5g sa networks,” *Journal of Network and Computer Applications*, vol. 221, p. 103765, 2024.
- [10] J. Groen, M. Belgiovine, U. Demir, B. Kim, and K. Chowdhury, “TRAC-TOR: Traffic analysis and classification tool for open ran,” in *ICC 2024 - IEEE International Conference on Communications*, 2024, pp. 4894–4899.
- [11] M. Wani, M. Kretschmer, B. Schröder, A. Grebe, and M. Rademacher, “Open RAN: A concise overview,” *IEEE Open Journal of the Communications Society*, vol. 6, pp. 13–28, 2025.
- [12] D. Anand, M. A. Togou, and G.-M. Muntean, “A machine learning-based xAPP for 5G O-RAN to mitigate co-tier interference and improve QoE for various services in a HetNet environment,” in *2023 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*, 2023, pp. 1–6.
- [13] D. Pliatsios, “Comparison of swarm intelligence methods for joint resource orchestration in open radio access network,” in *2024 14th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP)*, 2024, pp. 632–637.
- [14] Z. Zeng and H. Zhang, “Research on lightweight core network solutions for 5G private networks,” in *4th International Conference on Computing, Networks and Internet of Things (CNIOT)*, 2023, p. 349–355.
- [15] M. Motaung, N. Mthethwa, B. Nokane, and M. Masonta, “Comparative analysis of 5G SDR-based open-source software platforms: OpenAirInterface and srsRAN,” in *2025 IEEE 3rd Wireless Africa Conference (WAC)*, 2025, pp. 1–7.
- [16] O-RAN Software Community, “Projects — O-RAN Software Community,” 2025. [Online]. Available: <https://docs.o-ran-sc.org/en/latest/projects.html>
- [17] Ettus Research, “USRP B210 USB Software Defined Radio.” [Online]. Available: <https://www.ettus.com/all-products/ub210-kit/>
- [18] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, p. 785–794.